

Segment 3 (STACK)

1. What is polish notation?

♣ **Ans:**

Polish notation named after the polish mathematician Jan Lukasiewicz, refers to the notation in which the operator symbol is placed before its two operands. For example,

+AB -CD *EF /GH

The fundamental property of Polish notation is that the order in which the operations are to be performed is completely determined by the positions of the operators and operands in the expression. Accordingly, one never needs parentheses when writing expressions in polish notation.

Reverse Polish notation refers to the analogous notation in which the operator symbol is placed after its two operands:

AB+ CD- EF* GH/

2. Write the algorithm for evaluation of a Postfix expression.
3. Write the algorithm for transforming Infix Expression into Postfix Expression.

◆ **Translate/convert the following infix expressions into its equivalent postfix expression**

[Usually given for 3 to 4 marks]

1. $K + L - M * N + (O \wedge P) * W / U / V * T + Q$
2. $(A + B) * C - (D - E) * (F + G)$

♣ **Ans:**

Input	Stack	Output
(	(	
A	(	A
+	(+	
B	(+	AB
)		AB+
*	*	AB+
C	*	AB+C
-	-	AB+C*
(	-(	AB+C*
D	-(	AB+C*D
-	-(-	AB+C*D
E	-(-	AB+C*DE
)	-	AB+C*DE-
*	-*	AB+C*DE-
(	-*(	AB+C*DE-
F	-*(	AB+C*DE-F
+	-*(+	AB+C*DE-F
G	-*(+	AB+C*DE-FG
)	-*	AB+C*DE-FG+
)		AB+C*DE-FG+*-

3. $((A+B)/D) \uparrow ((E-F)*G)$

♣ Ans:

Input	Stack	Output
(	(	
(	((	
A	((	A
+	((+	
B	((+	AB
)	(	AB+
/	(/	AB+
D	(/	AB+D
)		AB+D/
^	^	AB+D/
(	^(	AB+D/
(	^((	AB+D/
E	^((	AB+D/E
-	^((-	AB+D/E
F	^((-	AB+D/EF
)	^(	AB+D/EF-
*	^(*	AB+D/EF-
G	^(*	AB+D/EF-G
)	^	AB+D/EF-G*
)		AB+D/EF-G*^

4. $A*(B-D)/E+F$

♣ Ans:

Input	Stack	Output
A		A
*	*	A
(	*(	A
B	*(	AB
-	*(-	AB
D	*(-	ABD
)	*	ABD-
/	/	ABD-*
E	/	ABD-*E
+	+	ABD-*E/
F	+	ABD-*E/F
)		ABD-*E/F+

5. $((A+B)*D) \uparrow (E-F)$

♣ Ans:

Input	Stack	Output
(	(	
(	((	
A	((	A
+	((+	A

B	((+	AB
)	(	AB+
*	(*	AB+
D	(*	AB+D
)		AB+D*
^	^	AB+D*
(	^(	AB+D*
E	^(	AB+D*E
-	^(-	AB+D*E
F	^(-	AB+D*EF
)	^	AB+D*EF-
)		AB+D*EF-^

◆ Evaluate each of the following postfix expression using the algorithm that you have studied:
[Usually given for 3 marks]

1. 20, 7, 3, -, /, 5, 2, 8, -, *, +

♣ Ans:

Input	Stack	Output
20	20	
7	20, 7	
3	20, 7, 3	
-	20, 4	7-3 = 4
/	5	20/4 = 5
5	5, 5	
2	5, 5, 2	
8	5, 5, 2, 8	
-	5, 5, -6	2-8 = -6
*	5, -30	5*-6 = -30
+	-25	5+(-30) = -25

2. 3, 5, +, 6, 4, -, *, 4, 1, -, 2, ^, +

♣ Ans:

Input	Stack	Output
3	3	
5	3, 5	
+	8	3+5 = 8
6	8, 6	
4	8, 6, 4	
-	8, 2	6-4 = 2
*	16	8*2 = 16
4	16, 4	
1	16, 4, 1	
-	16, 3	4-1=3
2	16, 3, 2	
^	16, 9	3^2 = 9
+	25	16+9 = 25

3. 12, 7, 3, -, /, 2, 1, 5, -, *, +
4. 12, 7, 3, -, /, 2, N, 5, +, *, +
5. N, 5, +, 6, 4, -, *, 4, 1, -, 2, ^, +
6. 3, 1, +, 2, ^, N, 4, -, 2, *, +, 5, -

◆ **Push and POP operations on Stack:** [Usually given for 3 marks]

Suppose the following stack of integers is in memory where STACK is allocated N=8 memory cells:

TOP=4 STACK: 227,123, N,583, __, __, __, __

Find the output of the following program segment:

1. Call POP (STACK, ITEMA).
 Call POP (STACK, ITEMB).
 Call POP (STACK, ITEMC).
 Call PUSH (STACK, ITEMA+140).
 Call PUSH (STACK, 190).
 Call PUSH (STACK, ITEMB+ ITEMC).
 Call PUSH (STACK, ITEMA+ITEMB)
2. Repeat while TOP! = 0
 Call POP (STACK, ITEM).
 Write: ITEM.
3. Exit.

(N is the last three digits of our ID, i.e. if your ID is C151216, then N=216).

Suppose the following stack of integers is in memory where STACK is allocated N=6 memory cells and initially STACK is empty (i.e., TOP=0)

Find the output of the following module:

1. Set AAA:=10 and BBB:=20
2. Call PUSH (STACK, AAA).
 Call PUSH (STACK, 5).
 Call PUSH (STACK, BB+3).
 Call PUSH (STACK, 15).
 Call PUSH (STACK, AAA+BBB).
3. Repeat while TOP! = 0
 Call POP (STACK, ITEM).
 Write: ITEM.
4. Return.

Suppose the following stack of integers is in memory where STACK is allocated N=6 memory cells:

TOP=3 STACK: 5,2, 3, __, __, __

Find the output of the following program segment:

4. Call POP (STACK, ITEMA).
 Call POP (STACK, ITEMB).
 Call POP (STACK, ITEMC).
 Call PUSH (STACK, ITEMB+2).
 Call PUSH (STACK, 8).
 Call PUSH (STACK, ITEMA+ITEMB)
5. Repeat while TOP! = 0
 Call POP (STACK, ITEM).
 Write: ITEM.
6. Exit.

Consider the following stack of city names:

STACK: London, Berlin, Rome, Paris, __, __

1. Describe the stack as the following operations take place:
 PUSH(STACK, Athens)

POP(STACK, ITEM)
 POP(STACK, ITEM)
 PUSH(STACK, Madrid)
 PUSH(STACK, Moscow)
 POP(STACK, ITEM)

2. Describe the stack if the operation POP(STACK, ITEM) deletes London

Given is the following stack of characters of 6 memory cells-

STACK: I, I, U, C, __, __

Describe the stack as the following operations take place-

- a. POP(STACK, ITEM)
- b. PUSH(STACK, C)
- c. PUSH(STACK, P)
- d. POP(STACK, ITEM)

◆ Theory questions: *[for 2 to 3 marks]*

1. What is stack? What are the operations are associated with stack? What are the applications of the stack?
2. What is Big O notation? What are best case, worst case, average case? *[2 marks]*
3. Write a procedure to PUSH an item onto a stack and POP an item from the stack.
4. What do you mean by *complexity* of an algorithm? Suppose, a problem can be solved in two different ways. The complexity of one algorithm is $C(n) = n^3$ and another is $C(n) = 2^n$. which algorithm will you choose and why? *[for 3 marks]*
5. Analyze the complexity of linear search algorithm.